

APPENDIX

This appendix collects the formal statements, proofs, and supporting derivations referenced in the main text, together with additional experimental tables and figures.

A. Setup and interventional means on mutilated graphs

With variables ordered topologically and the convention that W_{ij} is the coefficient of V_i in the equation for V_j , W is strictly upper triangular, so $W^\top$ is strictly lower triangular and nilpotent with $(W^\top)^p = 0$ [15, Thm. 3.1.5]. Hence $I - W^\top$ is invertible with the terminating Neumann series

$$(I - W^\top)^{-1} = I + W^\top + \dots + (W^\top)^{p-1}, \quad (\text{A.1})$$

and the reduced form is $V = (I - W^\top)^{-1}\varepsilon$.

For an intervention $do(X_s = x_s)$, incoming edges into X_s are deleted and X_s is fixed to x_s . Writing $R_s = \mathcal{V} \setminus X_s$ for the non-intervened variables, the restricted structural equations are $R_s = W_{R_s R_s}^\top R_s + W_{X_s R_s}^\top x_s + \varepsilon_{R_s}$, where $W_{X_s R_s}$ is the submatrix of W with rows in X_s and columns in R_s . Since $W_{R_s R_s}$ inherits strict triangularity, $I - W_{R_s R_s}^\top$ is invertible and

$$R_s = (I - W_{R_s R_s}^\top)^{-1} W_{X_s R_s}^\top x_s + (I - W_{R_s R_s}^\top)^{-1} \varepsilon_{R_s}. \quad (\text{A.2})$$

Taking expectations and using $\mathbb{E}[\varepsilon_{R_s}] = 0$ gives the interventional mean of $V_y \in R_s$ as the deterministic part, $f_s(x_s) = e_y^\top (I - W_{R_s R_s}^\top)^{-1} W_{X_s R_s}^\top x_s$. This step is valid even when Ω is non-diagonal: an intervention is not a conditioning event, so it leaves the disturbance distribution, and hence its mean, unchanged. Transposing the scalar right-hand side and using $(I - W_{R_s R_s}^\top)^{-\top} = (I - W_{R_s R_s})^{-1}$ yields the source-to-target form $f_s(x_s) = x_s^\top W_{X_s R_s} (I - W_{R_s R_s})^{-1} e_y$, i.e. (4) with $\tau_s(W) = W_{X_s R_s} (I - W_{R_s R_s})^{-1} e_y$. When interventions reach only direct parents, τ_s collects direct edge coefficients into V_y ; for deeper ancestors it accumulates products and sums of path coefficients, which is why θ_Y cannot in general be reduced to the parent coefficients of V_y .

B. Formal assumptions and their roles

Assumption 1 (Identifiability and constructive shared parameterization). For every $s \in \mathcal{E}$ the interventional mean $f_s(x_s) = \mathbb{E}[V_y \mid do(X_s = x_s)]$ is identifiable from $P(V)$ under the known graph (including latent confounding represented by correlated disturbances), and in the linear Gaussian SEM can be written $f_s(x_s) = g_s(x_s; \theta_Y)$ for a shared vector $\theta_Y \in \mathbb{R}^d$ of identifiable causal parameters or functionals sufficient to express $\{f_s : s \in \mathcal{E}\}$. The vector θ_Y is obtained constructively by applying the ID algorithm to each query $P(V_y \mid do(X_s = x_s))$ and collecting the unique identifiable quantities that appear [18, 23].

This ensures the optimization target is well defined from observational information; if some f_s were not identifiable, its prior mean and causal uncertainty could not be computed without extra assumptions or interventional data. The shared parameterization is what enables transfer: if two responses overlap in θ_Y , evidence for one reduces uncertainty for the other. Without it the method reverts to separate per-set surrogates.

Assumption 2 (Posterior regularity). Given observational data $D_{N_t}^O$, the posterior over the shared parameterization is approximately Gaussian, $\theta_Y \mid D_{N_t}^O \approx \mathcal{N}(\hat{\theta}_{Y,t}, \Sigma_{\theta,t})$, exactly so for conjugate Gaussian parameters or linear functionals of them, and otherwise via a delta-method, Laplace, or large-sample approximation. Here $\Sigma_{\theta,t}$ is the posterior covariance of the chosen θ_Y , not necessarily only of primitive edge coefficients.

This makes the cross-intervention covariance tractable: the kernel linearizes each g_s in the same posterior uncertainty. If the posterior is exactly Gaussian and g_s is linear, the covariance is exact; otherwise it is the usual first-order delta-method covariance with second-order error controlled in Theorem 1. Without it the kernel could still be estimated by Monte Carlo, but the closed-form finite-rank and information-gain statements would not hold exactly.

Assumption 3 (Bounded intervention domain and bounded causal variance). Each $\mathcal{X}_s$ is compact and there exists $\kappa < \infty$ with $k_{\text{causal}}(z, z) \leq \kappa^2$ for all $z = (s, x_s)$. A sufficient condition is compact domains, uniformly bounded Jacobians, and bounded $\|\Sigma_\theta\|$.

This is the standard bounded-variance condition for GP-UCB information-gain bounds and is natural for intervention design: feasible ranges are finite and smooth responses have bounded sensitivities on compact domains.

Assumption 4 (Fixed reference kernel). For the main regret theorem the kernel is built once from a fixed $\Sigma_{\theta,0}$ and linearization point $\hat{\theta}_{Y,0}$, i.e. $k_{\text{causal}}(z, z') = J_z(\hat{\theta}_{Y,0}) \Sigma_{\theta,0} J_{z'}(\hat{\theta}_{Y,0})^\top$, and is held fixed during the analysis.

This is technical rather than conceptual: classical GP-UCB regret bounds are stated for a fixed kernel and its maximum information gain [24]. Updating the kernel as data arrive makes the analysis a nonstationary-kernel problem; freezing it gives a conservative but clean object (Appendix K).

Remark 1 (Cost of the fixed-kernel assumption). Assumption 4 is conservative. With a small initial dataset, $\Sigma_{\theta,0}$ may be over-dispersed, so the bound uses the rank and information gain of this initial kernel; as N_t grows and $\Sigma_{\theta,t}$ shrinks, the effective rank of the adaptive kernel may fall and a time-varying analysis could be tighter.

C. Proof of Theorem 1 (cross-intervention covariance)

Let $\delta = \theta_Y - \hat{\theta}_Y$, so $\delta \sim \mathcal{N}(0, \Sigma_\theta)$ under Assumption 2. A first-order Taylor expansion gives $f_s(x_s) = g_s(x_s; \hat{\theta}_Y) + J_s(x_s)\delta + r_s(x_s)$ and likewise for f_t . The deterministic terms do not contribute to covariance, so

$$\text{Cov}(f_s, f_t | D^O) = \text{Cov}(J_s\delta, J_t\delta) + \text{Cov}(J_s\delta, r_t) + \text{Cov}(r_s, J_t\delta) + \text{Cov}(r_s, r_t). \quad (\text{A.3})$$

The leading term is $\text{Cov}(J_s\delta, J_t\delta) = J_s\Sigma_\theta J_t^\top$, which is (6). Each second-order remainder can be written $r_t = \frac{1}{2}\delta^\top H_t(\xi_t)\delta$, so the mixed terms involve third central moments of the centered Gaussian δ and vanish. For the remainder–remainder term, Taylor’s theorem with bounded Hessians gives $|r_s| \leq \frac{1}{2}H_s\|\delta\|^2$ and $|r_t| \leq \frac{1}{2}H_t\|\delta\|^2$, so by Cauchy–Schwarz $|\text{Cov}(r_s, r_t)| \leq \frac{1}{4}H_sH_t\mathbb{E}\|\delta\|^4$. The fourth-moment identity $\mathbb{E}\|\delta\|^4 = (\text{tr}\Sigma_\theta)^2 + 2\|\Sigma_\theta\|_F^2 \leq 3(\text{tr}\Sigma_\theta)^2$ then yields $|\text{Cov}(r_s, r_t)| \leq \frac{3}{4}H_sH_t(\text{tr}\Sigma_\theta)^2$, i.e. the omitted correction is $O((\text{tr}\Sigma_\theta)^2)$. If g_s, g_t are linear in θ_Y then $r_s = r_t = 0$ and (6) is exact. $\square$

Remark 2 (Beyond exact Gaussian posteriors). The mixed terms vanish because δ is centered Gaussian. Under a Laplace or large-sample approximation they need not vanish exactly, but under standard regularity they are higher order than the first-order covariance and can be absorbed into the causal-estimation error of the regret bound.

D. The coupled kernel: positive semidefiniteness, finite rank, and special cases

The kernel (7) is positive semidefinite via the feature map $\phi_s(x_s) = \Sigma_\theta^{1/2}J_s(x_s)^\top$, since for any queries z_i and coefficients c_i , $\sum_{i,j} c_i c_j k_{\text{causal}}(z_i, z_j) = \|\sum_i c_i \phi_{s_i}(x_{s_i})\|^2 \geq 0$. It is thus a finite-dimensional linear kernel over $\mathbb{R}^d$, $d = \dim(\theta_Y)$.

Proof of Corollary 1. Stacking features into $\Phi_T \in \mathbb{R}^{T \times d}$ gives $K_T = \Phi_T \Phi_T^\top$, so $\text{rank}(K_T) = \text{rank}(\Phi_T) \leq d = \dim(\theta_Y)$. Taking the supremum over finite query sets gives $r_{\text{causal}} \leq \dim(\theta_Y)$, and any single $\text{rank}(K_T) \leq r_{\text{causal}}$ by definition of the supremum. $\square$

Direct-parent interventions. If $X_s = \text{Pa}(V_y)$ then under $do(X_s = x_s)$ the equation for V_y gives $f_s(x_s) = \sum_{i \in \text{Pa}(V_y)} \beta_{iy} x_i$, so $\theta_Y = (\beta_{iy} : i \in \text{Pa}(V_y))$ and $r_{\text{causal}} \leq |\text{Pa}(V_y)|$. If $X_s \subsetneq \text{Pa}(V_y)$, then $f_s(x_s) = \sum_{i \in X_s} \beta_{iy} x_i + \sum_{j \in \text{Pa}(V_y) \setminus X_s} \beta_{jy} \mathbb{E}[V_j | do(X_s = x_s)]$, whose second term may introduce additional upstream parameters; the parent bound is exact only when all parents are intervened on or the remaining parental means add no new unknowns.

Chain example. For $X \rightarrow Z \rightarrow Y$ with $Z = aX + \varepsilon_Z$, $Y = bZ + \varepsilon_Y$, the effects are $f_X(x) = abx$ and $f_Z(z) = bz$, so $|\text{Pa}(Y)| = 1$ but $\theta_Y = (a, b)$ and $\dim(\theta_Y) = 2$. Thus $|\text{Pa}(V_y)|$ is not the right default complexity; the general bound is $r_{\text{causal}} \leq \dim(\theta_Y)$, with known coefficients reducing the dimension (e.g. if b is known, $\theta_Y = (a)$).

Remark 3 (Estimated Jacobians). In multiplicative cases the Jacobian depends on the unknown parameters; e.g. $J_X(x) = [bx \ a x]$ depends on (a, b) and is evaluated at $(\hat{a}, \hat{b})$, so k_{causal} is itself estimated. Assumption 4 absorbs this by fixing the kernel at the initial estimate; the extra error enters the causal-estimation term of Theorem 2.

Worked cross-covariance. The closed-form kernel for this chain—the cross-covariance (8) together with the two self-covariances—is given in the main text (Section III-E). The kernel is nonstationary in the intervention values, which motivates the fixed-reference choice of Assumption 4: the Jacobians are evaluated once at $\hat{\theta}_Y$ so that fixed-kernel GP-UCB bounds apply.

E. Information gain: proof of Lemma 1

For a GP with kernel matrix K_A and noise σ^2 , the mutual information between latent values and noisy observations is $I(y_A; f_A) = \frac{1}{2} \log \det(I + \sigma^{-2} K_A)$, and $\gamma_T = \max_{|A|=T} I(y_A; f_A)$ [24, 26]. Let $\lambda_1, \dots, \lambda_r$ be the nonzero eigenvalues of K_A , $r \leq r_{\text{causal}}$. Then $\log \det(I + \sigma^{-2} K_A) = \sum_{j=1}^r \log(1 + \sigma^{-2} \lambda_j)$, and by concavity of $\log(1 + \cdot)$ and Jensen, this is at most $r \log(1 + \frac{\sigma^{-2}}{r} \sum_j \lambda_j)$. Since $\sum_j \lambda_j = \text{tr}(K_A) = \sum_i k_{\text{causal}}(z_i, z_i) \leq T\kappa^2$, we get $I(y_A; f_A) \leq \frac{r}{2} \log(1 + \frac{T\kappa^2}{\sigma^2 r})$. The map $h(r) = \frac{r}{2} \log(1 + \frac{T\kappa^2}{\sigma^2 r})$ is increasing (using $\log(1 + u) \geq u/(1 + u)$), so replacing r by r_{causal} and maximizing over A gives (9); the big- O form follows from $r_{\text{causal}} \leq \dim(\theta_Y)$. $\square$

Remark 4 (Effective-rank refinement). The bound is conservative when Σ_θ has low effective rank: strongly identified parameters contribute negligibly, so one may replace r_{causal} by $r_{\text{eff}} = \#\{j : \lambda_j(\Sigma_\theta) \geq \delta\}$ plus a residual-trace term, giving a tighter bound when the spectrum decays quickly.

F. Causal-UCB acquisition: formal details

With $\hat{\sigma}_{t-1}^{\text{causal}}(z) = [J_z \Sigma_{\theta,t} J_z^\top]^{1/2}$ the posterior standard deviation of the first-order causal response, the practical acquisition is (10). For the regret analysis we separate the empirical causal bonus from the high-probability estimation error: assume with probability $\geq 1 - \delta$ a predictable sequence ϵ_t with $\sup_z |\hat{\mu}_{t-1}(z) - f(z)| \leq \epsilon_t$, and decompose $\epsilon_t = \epsilon_{\text{param}}(N_t, \delta) + \epsilon_{\text{adj}}(N_t, M_t, \delta)$, with $\epsilon_{\text{adj}} = 0$ in the closed-form linear Gaussian case. Under Assumption 3 the Jacobians

are uniformly bounded, $\sup_z \|J_z\| \leq L$, so by the mean-value theorem $\sup_z |g_z(\hat{\theta}_{Y,t}) - g_z(\theta_Y)| \leq L\|\hat{\theta}_{Y,t} - \theta_Y\|$ and we may take $\epsilon_{\text{param}}(N_t, \delta) = L\|\hat{\theta}_{Y,t} - \theta_Y\|$ on the corresponding event. The regret-safe acquisition is

$$\alpha_t^{\text{safe}}(z) = \hat{\mu}_{t-1}(z) + \sqrt{\beta_t} \hat{\sigma}_{t-1}^{\text{func}}(z) + \epsilon_t - \eta c(z); \quad (\text{A.4})$$

the additive ϵ_t avoids degeneracy where the Jacobian vanishes (e.g. $J_Z(0) = 0$ in the chain), whereas the multiplicative bonus $\lambda_t \hat{\sigma}_{t-1}^{\text{causal}}(z)$ remains useful in practice for steering exploration. A pointwise calibration $\lambda_t \hat{\sigma}_{t-1}^{\text{causal}}(z_t) \geq \epsilon_t(z_t)$ also suffices; we use (A.4) in the theorem for robustness to zero-Jacobian queries.

G. Proof of Theorem 2 (dynamic regret)

Define $f_{\text{all}}^* = \sup_{s, x_s} f_s(x_s)$, $f_{\mathcal{E}}^* = \sup_{s \in \mathcal{E}, x_s} f_s(x_s)$, $\epsilon_{\mathcal{E}} = f_{\text{all}}^* - f_{\mathcal{E}}^*$, and instantaneous regret $r_t = f_{\text{all}}^* - f(z_t)$. Decompose $r_t = \epsilon_{\mathcal{E}} + (f_{\mathcal{E}}^* - f(z_t))$, so $R_T = T\epsilon_{\mathcal{E}} + \sum_t (f_{\mathcal{E}}^* - f(z_t))$. On the causal-estimation event $\sup_z |\hat{\mu}_{t-1}(z) - f(z)| \leq \epsilon_t$ and the standard GP-UCB confidence event, $f(z) \leq \hat{\mu}_{t-1}(z) + \sqrt{\beta_t} \hat{\sigma}_{t-1}^{\text{func}}(z) + \epsilon_t$ and the matching lower bound hold for all z . Since z_t maximizes α_t^{safe} , with $z_t^* \in \arg \max_z f(z)$,

$$\begin{aligned} f(z_t^*) - f(z_t) &\leq \alpha_t^{\text{safe}}(z_t^*) + \eta c(z_t^*) - f(z_t) \leq \alpha_t^{\text{safe}}(z_t) + \eta c(z_t^*) - f(z_t) \\ &\leq 2\sqrt{\beta_t} \hat{\sigma}_{t-1}^{\text{func}}(z_t) + 2\epsilon_t + \eta(c(z_t^*) - c(z_t)). \end{aligned} \quad (\text{A.5})$$

For value regret we set $\eta = 0$ (or treat the cost term separately), giving $f(z_t^*) - f(z_t) \leq 2\sqrt{\beta_t} \hat{\sigma}_{t-1}^{\text{func}}(z_t) + 2\epsilon_t$. Summing and using the GP-UCB information-gain inequality $\sum_t \hat{\sigma}_{t-1}^{\text{func}}(z_t) \leq C' \sqrt{T \gamma_T^{\text{causal}}}$ [24, Lemma 5.4] gives $\sum_t (f_{\mathcal{E}}^* - f(z_t)) \leq C \sqrt{T \beta_T \gamma_T^{\text{causal}}} + 2 \sum_t \epsilon_t$. Substituting $\epsilon_t = \epsilon_{\text{param}} + \epsilon_{\text{adj}}$ and adding $T\epsilon_{\mathcal{E}}$ proves (11); Lemma 1 and Corollary 1 replace γ_T^{causal} by $O(r_{\text{causal}} \log T) \leq O(\dim(\theta_Y) \log T)$. The factor 2 is the usual upper/lower-confidence gap. $\square$

Remark 5 (Value vs. cost-adjusted regret). The theorem bounds value regret $f_{\text{all}}^* - f(z_t)$; the cost penalty $-\eta c(z)$ may still be used in the acquisition to discourage expensive interventions. A cost-adjusted bound follows by replacing f with $f - \eta c$ throughout.

H. Causal parameter concentration and growing data

When θ_Y consists of structural coefficients or linear functionals of them, each equation is estimated by Gaussian linear regression with posterior covariance $\Sigma_{\theta,t} = (\Sigma_{\theta,0}^{-1} + X_t^\top X_t / \sigma_\epsilon^2)^{-1}$, where X_t is the relevant design matrix. Under the regular-design condition $\lambda_{\min}(X_t^\top X_t) \asymp N_t$, $\|\Sigma_{\theta,t}\| = O(N_t^{-1})$ and standard concentration gives, with probability $\geq 1 - \delta$, $\|\hat{\theta}_{Y,t} - \theta_Y\| = O(\sqrt{(\dim(\theta_Y) + \log(1/\delta))/N_t})$. Combined with the Lipschitz bound $\sup_z |g_z(\hat{\theta}_{Y,t}) - g_z(\theta_Y)| \leq L\|\hat{\theta}_{Y,t} - \theta_Y\|$ this yields

$$\epsilon_{\text{param}}(N_t, \delta) = O\left(L \sqrt{\frac{\dim(\theta_Y) + \log(1/\delta)}{N_t}}\right). \quad (\text{A.6})$$

The same rate follows for smooth nonlinear functionals by the delta method under bounded functional Jacobians, with $\Sigma_{\theta,t}$ the covariance of the chosen parameterization [19].

Remark 6 (Weakly identified components). The regular-design condition assumes the data are informative about all of θ_Y . Under partial confounding or poor overlap, (A.6) applies only to the well-identified components and the analysis specializes to the effective rank of $\Sigma_{\theta,t}$ as in Remark 4.

Corollary 2 (Regret under growing observational data). *In the clean identifiable linear Gaussian setting with $\epsilon_{\text{adj}} = 0$, $\epsilon_{\mathcal{E}} = 0$, and $\epsilon_{\text{param}}(N_t, \delta) = O_p(N_t^{-1/2})$, if $N_t \asymp t^\alpha$ with $0 < \alpha \leq 1$ then $\sum_t \epsilon_{\text{param}} = O_p(T^{1-\alpha/2})$ and*

$$R_T = O(\sqrt{T \beta_T r_{\text{causal}} \log T}) + O_p(T^{1-\alpha/2}). \quad (\text{A.7})$$

In particular for $N_t \asymp t$ both terms scale as $\sqrt{T}$, and the causal-estimation term is asymptotically dominated by the coupled-UCB term whenever $\beta_T r_{\text{causal}} \log T \rightarrow \infty$.

Proof. With $N_t^{-1/2} \asymp t^{-\alpha/2}$, $\sum_{t=1}^T t^{-\alpha/2} = O(T^{1-\alpha/2})$ for $0 < \alpha \leq 1$; substituting into Theorem 2 gives (A.7), and $\alpha = 1$ gives the $\sqrt{T}$ statement. $\square$

I. Comparison with independent intervention-set GPs

Independent CBO [1] places a separate GP on each f_s . If T_s queries go to set s with $\sum_s T_s = T$, the block-diagonal joint kernel has information gain $\gamma_T^{\text{ind}} = \max_{\{T_s\}} \sum_s \gamma_{T_s}^{(s)} \leq \sum_s \gamma_T^{(s)}$. The coupled surrogate instead satisfies $\gamma_T^{\text{causal}} \leq O(r_{\text{causal}} \log T) \leq O(\dim(\theta_Y) \log T)$, replacing a sum of per-set terms by a single finite-rank term. Substituting into $R_T = O(\sqrt{T \beta_T \gamma_T})$, the advantage is largest when $r_{\text{causal}} \log T \ll \gamma_T^{\text{ind}}$, i.e. exactly when many interventional responses are different views of the same mechanisms. We deliberately avoid a kernel-free dimensional comparison such as $\sum_s \dim(X_s)$ versus $\dim(\theta_Y)$, since squared-exponential, Matérn, and finite-rank linear kernels have different information-gain rates; the clean

comparison is in information gain. In the chain with $\mathcal{E} = \{\{X\}, \{Z\}\}$, the independent surrogate uses two separate one-dimensional GPs and cannot transfer between f_X and f_Z despite their shared b , whereas the coupled surrogate connects them through $k_{\text{causal}}((X, x), (Z, z)) = J_X(x) \Sigma_\theta J_Z(z)^\top$.

J. Nonlinear extension: path-Lipschitz propagation

For nonlinear SEMs $V_i = g_i(\text{Pa}_i) + \varepsilon_i$ with $g_i \sim \mathcal{GP}$, intervening mutilates the graph and uncertainty is propagated through the remaining DAG (by Monte Carlo, moment matching, sigma points, or Bayesian quadrature). Assume each mechanism is coordinate-wise Lipschitz, $|g_j(u) - g_j(u')| \leq \sum_{i \in \text{Pa}(j)} L_{ij} |u_i - u'_i|$, and learned to accuracy $\sup_u |\hat{g}_i(u) - g_i(u)| \leq \epsilon_i$ on a compact domain (with $\epsilon_i = O_p(N_i^{-\beta_i})$ for GP mechanisms). For a directed path $\pi : i \rightsquigarrow y$ define its sensitivity $L(\pi) = \prod_{(a \rightarrow b) \in \pi} L_{ab}$, and the total sensitivity from V_i to V_y in the mutilated graph $G_{\bar{X}_s}^{(s)}$ as $L_{i \rightarrow y}^{(s)} = \sum_{\pi: i \rightsquigarrow y} L(\pi)$.

Proposition 1 (Path-Lipschitz nonlinear error propagation). *Under the conditions above, $|\hat{f}_s(x_s) - f_s(x_s)| \leq \sum_{i \in \text{An}(y; G_{\bar{X}_s})} L_{i \rightarrow y}^{(s)} \epsilon_i$.*

Proof. In topological order on $G_{\bar{X}_s}$, for any non-intervened j , $|\hat{V}_j - V_j| \leq |\hat{g}_j(\hat{\text{Pa}}_j) - g_j(\hat{\text{Pa}}_j)| + |g_j(\hat{\text{Pa}}_j) - g_j(\text{Pa}_j)| \leq \epsilon_j + \sum_{i \in \text{Pa}(j)} L_{ij} |\hat{V}_i - V_i|$. Unrolling along directed paths to V_y gives $|\hat{V}_y - V_y| \leq \sum_{i \in \text{An}(y)} (\sum_{\pi: i \rightsquigarrow y} \prod_{(a \rightarrow b) \in \pi} L_{ab}) \epsilon_i = \sum_i L_{i \rightarrow y}^{(s)} \epsilon_i$ pointwise. Taking the intervened expectation and applying Jensen's inequality gives the claim. $\square$

Remark 7. Errors are amplified multiplicatively along paths to V_y ; long high-sensitivity paths amplify uncertainty more than shallow or weakly-coupled graphs. The bound applies Jensen to the absolute pointwise error and so may be conservative when errors partially cancel under the intervened distribution.

K. Adaptive-kernel extension

The empirically natural version updates $\hat{\theta}_{Y,t}, \Sigma_{\theta,t}$ and re-evaluates the Jacobians, giving the adaptive kernel $k_t(z, z') = J_z(\hat{\theta}_{Y,t}) \Sigma_{\theta,t} J_{z'}(\hat{\theta}_{Y,t})^\top$. Because the kernel changes over time, fixed-kernel GP-UCB theory does not apply directly; we add two ingredients.

Assumption 5 (Adaptive-kernel confidence with drift). At each t , with probability $\geq 1 - \delta$, $|f(z) - \hat{\mu}_{t-1}^{(t)}(z)| \leq \sqrt{\beta_t} \hat{\sigma}_{t-1}^{(t)}(z) + \epsilon_t + \rho_t$ for all z , where $\hat{\mu}^{(t)}, \hat{\sigma}^{(t)}$ use k_t and $\rho_t \geq 0$ is a kernel-drift error.

Remark 8 (Interpretation of the drift term). ρ_t bounds the error of replacing a fixed reference kernel by the estimated k_t , absorbing changes in linearization point, covariance, and induced posterior. If the Jacobians are Lipschitz in θ_Y , the domain is compact, and the covariance changes smoothly, $\rho_t \lesssim L_J \|\hat{\theta}_{Y,t} - \hat{\theta}_{Y,t-1}\| \|\Sigma_{\theta,t}\| + \|\Sigma_{\theta,t} - \Sigma_{\theta,t-1}\|$ up to constants.

Assumption 6 (Adaptive variance sum). With $\bar{\gamma}_T = \sup_{1 \leq t \leq T} \gamma_T(k_t)$, $\sum_{t=1}^T (\hat{\sigma}_{t-1}^{(t)}(z_t))^2 \leq C_\gamma \bar{\gamma}_T$ for some $C_\gamma > 0$.

This is an explicit slow-drift condition: the fixed-kernel variance-sum inequality [24, Lemma 5.3] needs the same kernel throughout, so for a time-varying kernel we require the variance sum to remain controlled; it is expected when $\|k_t - k_{t-1}\|_\infty$ decays.

Theorem 3 (Approximate regret for adaptive GC-CBO). *Under Assumptions 1–3, 5, and 6, the adaptive causal-UCB rule satisfies*

$$R_T \leq C \sqrt{T \beta_T \bar{\gamma}_T} + 2 \sum_t (\epsilon_{\text{param}} + \epsilon_{\text{adj}}) + 2 \sum_t \rho_t + T \epsilon_\mathcal{E}. \quad (\text{A.8})$$

If each k_t has rank $\leq r_{\text{causal}}$ and $k_t(z, z) \leq \kappa^2$, then $\bar{\gamma}_T \leq \frac{r_{\text{causal}}}{2} \log(1 + \frac{T \kappa^2}{\sigma^2 r_{\text{causal}}})$ and the first term becomes $C \sqrt{T \beta_T r_{\text{causal}}} \log T$.

Proof. On the event of Assumption 5, the upper/lower-confidence argument of Appendix G, now carrying ρ_t , gives $f(z_t^*) - f(z_t) \leq 2\sqrt{\beta_t} \hat{\sigma}_{t-1}^{(t)}(z_t) + 2\epsilon_t + 2\rho_t$. Summing, applying Cauchy–Schwarz with $\beta_t \leq \beta_T$ and Assumption 6 gives $\sum_t \sqrt{\beta_t} \hat{\sigma}_{t-1}^{(t)}(z_t) \leq C' \sqrt{T \beta_T \bar{\gamma}_T}$, and adding $T \epsilon_\mathcal{E}$ and the estimation terms yields (A.8). Applying Lemma 1 to each k_t and taking the supremum bounds $\bar{\gamma}_T$. $\square$

Remark 9 (Comparison with the fixed-reference theorem). Relative to Theorem 2, the adaptive bound adds $2 \sum_t \rho_t$ and needs Assumption 6. Under slow drift with $\rho_t = O(t^{-1/2})$ the drift contribution is $O(\sqrt{T})$, matching the causal-estimation term under linear observational growth and not dominating the coupled-UCB term up to logarithmic and rank factors.

L. Soundness and complexity

Proposition 2 (Soundness of fixed-reference GC-CBO). *Under Assumptions 1–4, running Algorithm 1 in its fixed-reference variant with $\eta = 0$ on the causal-estimation and GP-UCB confidence events yields the regret bound (11). If additionally*

$\epsilon_{\text{adj}} = 0$, $\epsilon_{\mathcal{E}} = 0$, and $N_t \asymp t$, then $R_T/T = O(\sqrt{\beta_T \dim(\theta_Y) \log T/T}) + O_p(T^{-1/2})$, and the best true queried intervention satisfies $f_{\text{all}}^* - \max_{t \leq T} f(z_t) \leq R_T/T$.

Proof. The fixed-reference variant uses k_{causal} and the regret-safe acquisition analyzed in Theorem 2, so (11) follows with Corollary 1; Corollary 2 gives the averaged rate, and $f_{\text{all}}^* - \max_t f(z_t) \leq \frac{1}{T} \sum_t (f_{\text{all}}^* - f(z_t)) = R_T/T$ gives the simple-regret claim. $\square$

Complexity. Let $d = \dim(\theta_Y)$, $r = r_{\text{causal}} \leq d$, and $M = |\mathcal{Z}|$ the number of candidate queries after discretization. The observational posterior update costs $O(d^3)$ (or $O(d^2)$ with online updates). Because $K_t = \Phi_t \Phi_t^\top$ with $\Phi_t \in \mathbb{R}^{t \times r}$, the matrix-inversion lemma gives an $O(tr^2 + r^3)$ GP solve and $O(tr + r^2)$ memory; acquisition over M candidates costs about $O(Mr^2)$ per sweep. Over T rounds the total is $O(T^2r^2 + Tr^3 + TMr^2)$ with $O(tr + d^2)$ memory, much cheaper than a dense GP when $r \ll T$. Against independent CBO, which recomputes separate factorizations at cost $\sum_s O(t_s^3)$ and never shares across sets, the coupled surrogate’s main advantage is statistical: interventional data for one set reduce uncertainty for another whenever responses share identifiable parameters (Appendix I).

M. Additional experimental details, tables, and figures

This section reports the detailed no-parent ECOLI70 study and the per-variant ablation that the main text summarizes. Tables report terminal mean, standard deviation, and median, with cumulative cost and the first cost at which the median reaches an ϵ -neighborhood of the optimum where applicable. The theory diagnostics (Table II), the transfer stress-test sweep (Table III), the benchmark convergence curves (Figure 3), and the no-parent ECOLI70 sweep and ablation plots (Figure 4) appear in the main text.

As in the main experiments, the horizontal axis in the ECOLI70 no-parent plots is cumulative *intervention* cost, not iteration count. Observational samples are used offline to estimate the initial causal model and are not charged on this axis. An intervention on a set S costs $|S|$, so singleton, pair, triple, and five-variable interventions cost 1, 2, 3, and 5, respectively. This convention is important for interpreting Figure 4: increasing the maximum intervention-set size both enlarges the admissible family of intervention sets and allows the optimizer to choose more expensive interventions. Thus, the cumulative cost naturally increases when moving from maximum set size 2 to 3 and then to 5.

The no-parent ECOLI70 study for target b1583 is broken out by maximum set size in Tables VI–V. For singleton and pair interventions the optimum is $f^* = 0.8427$ at `lacY;eutG`, and both CBO and GC-CBO reach terminal median 0.8427 (terminal mean 0.8930 due to seed variability). When triple interventions are allowed the optimum becomes 0.6422 at `lacY;eutG;fixC`; the sweep in Table VI shows GC-CBO improving the terminal mean from 0.7315 to 0.7154 and reducing the standard deviation from 0.1377 to 0.1097. When the maximum set size is increased to 5, the feasible optimum improves further to 0.3183, achieved by the five-variable ancestor intervention `lacY;eutG;fixC;cspG;sucA`. This is the hardest no-parent setting because the search space is much larger and the best feasible intervention is no longer contained in the pair or triple families. In this setting, independent CBO does not reach the optimum within the tested budget: its terminal mean is 0.7688 and its terminal median is 0.6927. GC-CBO, by contrast, reaches terminal mean 0.3183 and median 0.3183, matching the optimum with zero terminal variance across seeds. This is the regime the method targets: many related ancestor-intervention functions must be learned from sparse data, and the finite-rank coupling lets evidence from one intervention family inform another.

TABLE VI
ECOLI70 NO-PARENT INTERVENTION SWEEP FOR TARGET b1583. DIRECT PARENTS ARE EXCLUDED; LOWER IS BETTER. THE MAXIMUM SET SIZE CONTROLS THE LARGEST NUMBER OF VARIABLES THAT MAY BE JOINTLY MANIPULATED.

Size	Method	Mean	Std.	Median	f^*	Best set	Rank/dim.
2	CBO	0.8930	0.0872	0.8427	0.8427	<code>lacY;eutG</code>	–
2	GC-CBO	0.8930	0.0872	0.8427	0.8427	<code>lacY;eutG</code>	19/19
3	CBO	0.7315	0.1377	0.6571	0.6422	<code>lacY;eutG;fixC</code>	–
3	GC-CBO	0.7154	0.1097	0.6571	0.6422	<code>lacY;eutG;fixC</code>	19/19
5	CBO	0.7688	0.1723	0.6927	0.3183	<code>lacY;eutG;fixC;asnA</code>	–
5	GC-CBO	0.3183	0.0000	0.3183	0.3183	<code>lacY;eutG;fixC;cspG;sucA</code>	19/19

The sweep in Table VI also explains the qualitative behavior of the left panel of Figure 4. For maximum set sizes 2 and 3, both methods reach or approach the feasible optimum because the admissible intervention family is still moderate and the best set is relatively easy to discover. At size 2, the best feasible set is the pair `lacY;eutG`; at size 3, the best feasible set is the triple `lacY;eutG;fixC`. In both cases, independent CBO can still collect enough evidence about the relevant set-specific response surfaces. At size 5, however, the optimum moves to a higher-order intervention involving five non-parent ancestors. The number of admissible intervention sets grows rapidly, and CBO treats these response surfaces largely independently. As a result, CBO can find a good ancestor set but does not reliably discover the best five-variable intervention within the same

budget. GC-CBO uses the shared downstream Gaussian mechanism to transfer information across related ancestor-intervention functions, which is why it reaches the new optimum.

The ablation in Table VII isolates the causal prior and coupling effects in the no-parent max-set-size-5 setting. Removing the causal prior leaves CBO far from the optimum, with mean 1.2752 and median 1.3074. CBO with the prior improves to 0.7775, but still does not reach the optimum. GC-CBO without explicit coupling improves substantially over CBO, reaching mean 0.4315 and median 0.3183, but with nonzero variance. The EI variant is weaker in this sparse combinatorial setting, with mean 0.7934. The UCB and fixed-kernel GC-CBO variants are strongest: both reach mean, median, and optimum value 0.3183 with zero terminal variance across seeds. This supports the design used in the main experiments: the causal prior is essential, and the paper-aligned confidence-bound version of GC-CBO is best suited to the no-parent ancestor-search regime.

TABLE VII

ABLATION ON ECOLI70 NO-PARENT MAX-SET-SIZE-5. THE CAUSAL PRIOR IS ESSENTIAL; THE UCB AND FIXED-KERNEL GC-CBO VARIANTS REACH THE OPTIMUM. LOWER IS BETTER.

Method	Mean	Std.	Median	Cost to ϵ -opt.
CBO	0.7775	0.1363	0.7495	–
CBO-noprior	1.2752	0.1542	1.3074	–
GC-CBO no coupling	0.4315	0.1961	0.3183	28.3
GC-CBO EI	0.7934	0.1227	0.7734	–
GC-CBO UCB	0.3183	0.0000	0.3183	33.3
GC-CBO fixed kernel	0.3183	0.0000	0.3183	33.3